## Conservation Genetics Practice Problems #5

1) What will the equilibrium allele frequencies be at a biallelic locus with a mutation rate of $\mu = 5 * 10^{-7}$ from $A_1$ to $A_2$ and $v = 1 * 10^{-8}$ from $A_2$ to $A_1$?

2) Assuming a per-base mutation rate of $1 * 10^{-7}$, what is the probability there will be no new single-nucleotide mutations in a new generation of 20 diploid organisms with 100,000 base genomes?

3) Solve for $\mu$ if the frequency of a recessive deleterious allele ($s = 0.1$) at mutation-selection equilibrium is 0.15.

4) Assuming a one-way input of migrants at rate $m$ with a recessive allele at frequency $q_m$ into a focal population where the allele is originally found at a frequency $q_0$, what will happen to the per-generation change in the frequency of $q$ (i.e., $\Delta q$) over time?

   a) it will stay constant, because $m$ does not change
   b) it will increase, as more copies of $q$ are found in the population
   c) it will decline, as the frequencies of $q$ in the migrant and focal popualtion become more similar
   d) there is not enough information to say

5) What evolutionary processes produce a cline in allele frequencies, and why?